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Secure and Scalable Network Using Cisco Packet Tracer

Project Report
Course: Computer Networking (CSE 318)

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Contents

0.1	Abstract	3
0.2	Introduction	3
0.3	Objectives	3
0.4	Methodology	3
0.5	Network Architecture Design	4
0.5.1	Design Overview	4
0.5.2	Core Layer	4
0.5.3	Distribution Layer	4
0.5.4	Access Layer	4
0.5.5	Hierarchical Design Benefits	5
0.5.6	Logical Network Flow	5
0.5.7	Integration with VLANs and Routing	5
0.5.8	Design Justification	5
0.6	VLAN Configuration	6
0.7	IP Addressing Scheme	6
0.7.1	Subnet Design Strategy	6
0.7.2	IP Address Allocation Table	6
0.7.3	Default Gateway Configuration	7
0.7.4	Design Justification	7
0.8	Routing Implementation	7
0.9	Network Devices Used	7
0.9.1	Routers	7
0.9.2	Layer 3 Switches	8
0.9.3	Layer 2 Switches	8
0.9.4	Access Points	8
0.9.5	End Devices	8
0.9.6	Networking Cables and Interfaces	8

0.9.7	Device Role Summary	8
0.10	Server Implementation	9
0.10.1	Server Architecture	9
0.10.2	DNS Server	9
0.10.3	Web Server	9
0.10.4	File Server	9
0.10.5	Server VLAN Configuration	9
0.10.6	Security and Access Control	9
0.10.7	Benefits of Centralized Server System	10
0.11	Security Measures	10
0.12	Testing and Results	10
0.13	Advantages	10
0.14	Limitations	10
0.15	Future Improvements	11
0.16	Complex Engineering Problem (WP1–WP7)	11
0.17	Complex Engineering Problem (EA1 - EA2)	11
0.18	Conclusion	12
0.19	References	12

0.1 Abstract

This project presents the design and implementation of a secure and scalable enterprise network using Cisco Packet Tracer. The network follows a three-tier architecture consisting of Core, Distribution, and Access layers. Multiple departments are segmented using VLANs to enhance security and reduce broadcast traffic. Inter-VLAN routing is implemented to allow communication between departments. The design ensures efficient data transmission, centralized services, and future scalability.

0.2 Introduction

Modern organizations require efficient and secure communication systems to manage operations across multiple departments. Traditional flat networks often suffer from congestion, security issues, and poor scalability.

This project aims to design a structured enterprise network that improves performance, enhances security, supports scalability, and enables centralized management.

0.3 Objectives

- Design a hierarchical network using 3-layer architecture
- Implement VLAN-based segmentation
- Enable inter-department communication using routing
- Provide centralized server access
- Ensure network scalability and security

0.4 Methodology

The project is developed using Cisco Packet Tracer simulation:

- Network topology design
- IP addressing and VLAN planning
- Device configuration
- Routing implementation
- Testing and validation

0.5 Network Architecture Design

The proposed enterprise network is designed based on a scalable three-tier hierarchical architecture model, which is widely used in modern organizational networks to ensure performance, security, and manageability. The architecture is divided into three logical layers: Core, Distribution, and Access layers.

0.5.1 Design Overview

The overall network design follows a modular approach where each layer has a specific role. This separation of responsibilities improves fault isolation, reduces complexity, and enhances network performance. The architecture supports multiple departments through VLAN segmentation and enables secure inter-department communication through controlled routing.

0.5.2 Core Layer

The Core Layer serves as the backbone of the network. It is responsible for high-speed data transfer between different parts of the network with minimal latency. This layer is optimized for performance and reliability rather than complex processing or filtering. Redundancy and fast switching mechanisms are typically used in this layer to ensure continuous network availability.

0.5.3 Distribution Layer

The Distribution Layer acts as an intermediary between the Core and Access layers. It is responsible for:

- Inter-VLAN routing
- Policy enforcement and access control
- Packet filtering and traffic management
- Routing decisions based on network policies

This layer ensures that only authorized and properly routed traffic is forwarded between different VLANs and network segments.

0.5.4 Access Layer

The Access Layer is the point of entry for end-user devices. It connects PCs, laptops, printers, and wireless devices to the network. This layer is responsible for:

- Providing network access to end devices

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- VLAN assignment and segmentation
 - Basic security control such as port security

0.5.5 Hierarchical Design Benefits

The hierarchical architecture provides several advantages:

- **Scalability:** New departments or devices can be added without redesigning the entire network.
- **Performance Optimization:** Traffic is efficiently managed by separating roles across layers.
- **Fault Isolation:** Failures in one layer do not significantly impact the entire network.
- **Security Enhancement:** VLAN segmentation and controlled routing improve data security.
- **Manageability:** Easier configuration, monitoring, and troubleshooting.

0.5.6 Logical Network Flow

Data flow in the network follows a structured path:

1. End devices send data through the Access Layer.
2. The Distribution Layer processes routing and applies policies.
3. The Core Layer ensures fast backbone communication between network segments.
4. Responses follow the reverse path back to the destination device.

0.5.7 Integration with VLANs and Routing

Each department is assigned a dedicated VLAN to ensure logical separation of traffic. Inter-VLAN routing is performed at the Distribution Layer using Layer 3 switching. This ensures secure and controlled communication between departments such as Sales, HR, Finance, and IT.

0.5.8 Design Justification

This architecture is chosen because it provides a balance between simplicity and enterprise-level performance. It is widely used in real-world organizational networks due to its efficiency, modularity, and ability to scale with business growth.

0.6 VLAN Configuration

- VLAN 10 – Sales
- VLAN 20 – HR
- VLAN 30 – Finance
- VLAN 40 – Admin
- VLAN 50 – IT

0.7 IP Addressing Scheme

A structured IP addressing scheme is designed to ensure efficient network management, easy troubleshooting, and clear segmentation between different departments. Each VLAN is assigned a unique Class C subnet to logically separate network traffic and improve security.

This hierarchical addressing approach supports scalability and simplifies future expansion of the network.

0.7.1 Subnet Design Strategy

The IP addressing scheme follows a consistent pattern:

- Each department is assigned a dedicated VLAN.
- Each VLAN is mapped to a separate /24 subnet.
- The default gateway for each subnet is configured on the Layer 3 switch.
- Server network is isolated in a separate VLAN for security.

0.7.2 IP Address Allocation Table

VLAN ID	Department	Network Address	Usable IP Range
10	Sales & Marketing	192.168.10.0/24	192.168.10.1 – 192.168.10.254
20	Human Resources	192.168.20.0/24	192.168.20.1 – 192.168.20.254
30	Finance & Accounts	192.168.30.0/24	192.168.30.1 – 192.168.30.254
40	Administration	192.168.40.0/24	192.168.40.1 – 192.168.40.254
50	IT Department	192.168.50.0/24	192.168.50.1 – 192.168.50.254
60	Server Network	192.168.60.0/24	192.168.60.1 – 192.168.60.254

0.7.3 Default Gateway Configuration

Each VLAN is assigned a default gateway configured on the Layer 3 switch as follows:

- VLAN 10 Gateway: 192.168.10.1
- VLAN 20 Gateway: 192.168.20.1
- VLAN 30 Gateway: 192.168.30.1
- VLAN 40 Gateway: 192.168.40.1
- VLAN 50 Gateway: 192.168.50.1
- VLAN 60 Gateway: 192.168.60.1

0.7.4 Design Justification

This IP addressing scheme is designed to:

- Ensure logical separation of departments
- Reduce IP conflicts and configuration errors
- Improve scalability for future departments
- Simplify routing and network management
- Enhance security through network segmentation

0.8 Routing Implementation

Inter-VLAN routing is implemented using Layer 3 switches with SVIs acting as gateways. This allows controlled communication between VLANs while maintaining segmentation.

0.9 Network Devices Used

The proposed enterprise network utilizes a variety of networking devices to ensure efficient communication, scalability, and secure data transmission across all departments. Each device plays a specific role within the hierarchical architecture.

0.9.1 Routers

Routers are used to connect different networks and enable communication between external and internal networks. They are responsible for forwarding packets based on IP addresses and ensuring optimal routing paths when required.

0.9.2 Layer 3 Switches

Layer 3 switches perform both switching and routing functions. In this project, they are primarily used for inter-VLAN routing through SVIs (Switch Virtual Interfaces), enabling communication between different departmental VLANs.

0.9.3 Layer 2 Switches

Layer 2 switches are used in the Access Layer to connect end-user devices. They operate based on MAC addresses and are responsible for forwarding data within the same VLAN efficiently.

0.9.4 Access Points

Access points provide wireless connectivity for mobile devices such as laptops and smartphones. They extend network access without requiring physical cable connections.

0.9.5 End Devices

End devices include desktop computers, laptops, and printers used by different departments. These devices are assigned IP addresses based on their respective VLANs.

0.9.6 Networking Cables and Interfaces

Ethernet cables and switch interfaces are used to physically connect devices within the network. Proper cabling ensures reliable and high-speed data transmission.

0.9.7 Device Role Summary

Each device contributes to the overall performance of the network:

- Routers manage external connectivity
- Layer 3 switches handle inter-VLAN routing
- Layer 2 switches manage local traffic
- Access points enable wireless communication
- End devices generate and consume network data

0.10 Server Implementation

The server infrastructure is designed to provide centralized network services to all departments. All servers are placed in a dedicated VLAN to enhance security, control access, and ensure efficient resource management.

0.10.1 Server Architecture

A centralized server model is implemented where all services are hosted in a dedicated server network (VLAN 60). This design ensures better control, security, and scalability.

0.10.2 DNS Server

The DNS (Domain Name System) server is responsible for translating domain names into IP addresses. This allows users to access services using human-readable names instead of numeric IP addresses.

0.10.3 Web Server

The web server hosts internal web applications and organizational websites. It enables departments to access centralized web-based services through the network.

0.10.4 File Server

The file server provides centralized storage for documents, reports, and shared resources. It allows authorized users from different departments to upload, download, and manage files efficiently.

0.10.5 Server VLAN Configuration

All servers are placed in VLAN 60 with the following configuration:

- Network Address: 192.168.60.0/24
- Default Gateway: 192.168.60.1
- IP Range: 192.168.60.1 – 192.168.60.254

0.10.6 Security and Access Control

To ensure data protection, server access is restricted using VLAN segmentation and Access Control Lists (ACLs). Only authorized departments, such as the IT department, are allowed direct or controlled access to critical server resources.

0.10.7 Benefits of Centralized Server System

- Efficient resource sharing across departments
- Improved data security and access control
- Simplified network management
- Centralized backup and maintenance
- Better performance and scalability

0.11 Security Measures

- VLAN segmentation
- ACL-based control
- Restricted server access

0.12 Testing and Results

Ping tests confirmed successful intra-VLAN and inter-VLAN communication.

0.13 Advantages

- Scalable architecture
- High security
- Efficient traffic management
- Centralized services

0.14 Limitations

- No physical deployment
- Limited redundancy
- Basic simulation environment

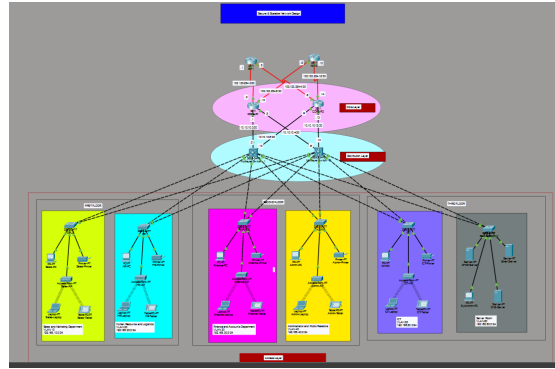


Figure 1: Networks Design

0.15 Future Improvements

- OSPF dynamic routing
- Firewall integration
- Redundant links
- Real hardware deployment

0.16 Complex Engineering Problem (WP1–WP7)

WP1 – Knowledge Depth: Requires VLANs, routing, subnetting, OSI model understanding.

WP2 – Conflicting Requirements: Balancing security, cost, and performance.

WP3 – Analysis Depth: Traffic flow and routing optimization.

WP4 – Familiarity: Enterprise-level networking design.

WP5 – Standards: IEEE 802.1Q and IP standards.

WP6 – Stakeholders: Multiple departments with different needs.

WP7 – Interdependence: All network components are interlinked.

0.17 Complex Engineering Problem (EA1 - EA2)

EA1 – Resources: Cisco Packet Tracer, routers, switches, servers, end devices, and networking protocols.

EA2 – Interactions: VLANs, routing protocols, switches, and servers working together for data communication.

EA3 – Innovation: Hierarchical network design, VLAN segmentation, and centralized server architecture.

EA4 – Impact: Network design affects security, reliability, and organizational performance in real systems.

EA5 – Familiarity: Application of enterprise-level networking concepts beyond basic network design.

0.18 Conclusion

This project successfully demonstrates a secure and scalable enterprise network using Cisco Packet Tracer. The design ensures efficient communication, security, and scalability for modern organizational requirements.

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